

When Does Communication Avoidance Pay?

Communication-Avoiding Krylov Exponential Integration for Multi-GPU Option Pricing

Kevin Knights | Supervisor: Prof. Kirk M. Soodhalter | School of Mathematics, Trinity College Dublin



1 The operator three assets, one exponential action

$$\frac{\partial u}{\partial \tau} = \frac{1}{2} \sum_{d,d'} \rho_{dd'} \sigma_d \sigma_{d'} \frac{\partial^2 u}{\partial x_d \partial x_{d'}} + \sum_d \left(r - \frac{1}{2} \sigma_d^2 \right) \frac{\partial u}{\partial x_d} - \mathbf{r}u + b(\tau)$$

■ Diffusion ■ Convection ■ Reaction ■ Forcing

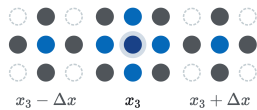
$$\tilde{A} = \begin{bmatrix} A & B \\ 0 & K \end{bmatrix}, \quad \tilde{v} = \begin{bmatrix} u_k \\ s(\tau_k) \end{bmatrix}, \quad s(\tau) = [\tau^2/2, \tau, 1]^T, \quad M = N + 3$$

$$u_{k+1} = e^{h\tilde{A}} \tilde{v}$$

Central differences give a non-symmetric A , $N = n^3$ rows; one action per step in $\mathcal{K}_m(\tilde{A}, \tilde{v})$, $m \leq m_{\max} = 39$ [1, 4].

19 nonzeros per row

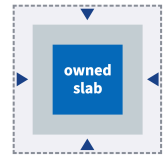
7-point stencil plus 3×4 mixed-derivative diagonals.



- center node
- 8 corners omitted
- 6 faces, diffusion
- 12 diagonals, correlation

The price of reuse

One exchange instead of s , paid for in recomputed boundary layers [3].



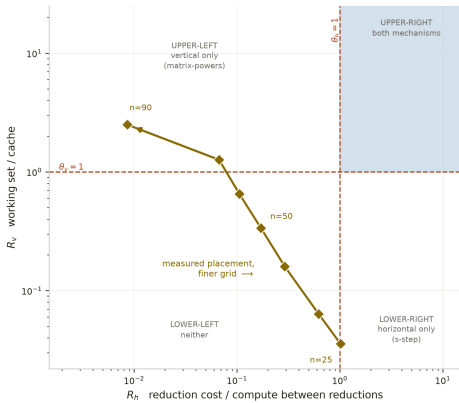
2 Predict, then measure two ratios from the operator and calibrated machine constants; the unit lines are equality lines, not crossovers

$$R_v = \frac{\text{Arnoldi working set}}{\text{last-level cache}}$$

Above 1: **matrix-powers reuse** is worth testing [2].

CPU: the corner is never reached

Refining the grid moves the operator past the corner, never into it.



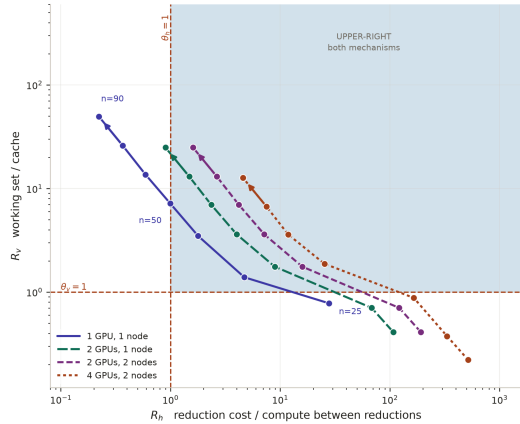
puffin, 24-core CPU node, single NUMA node. Eight grids, $n = 25 \rightarrow 90$, real Basket operator.

$$R_h = \frac{\text{reduction latency}}{\text{compute between them}}$$

Above 1: **grouped orthogonalization** is worth testing [2].

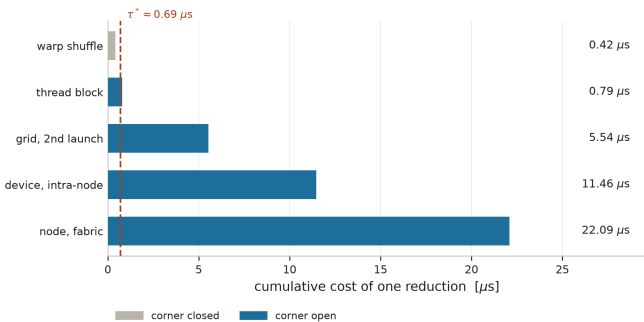
GPUs: inside the corner already

Identical axes, recalibrated constants. More devices push further right.

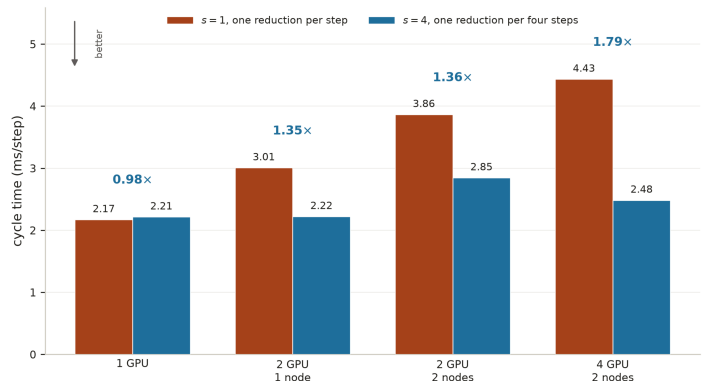
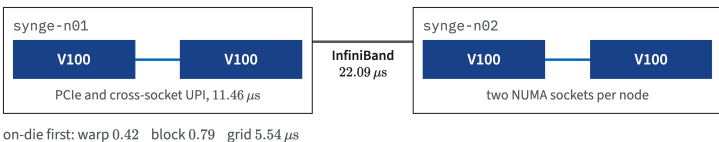


syngé, 1 to 4 Tesla V100 across two nodes, one grid per mark.

3 What it costs, what it bought one reduction across five levels of syngé, against equal-work controls at $n = 61$, $m = 39$



syngé: two nodes, two V100 each



Basket payoff, $n = 61$, $m = 39$, 100 steps, exact-depth halo so both widths do equal numerical work.

11-16×
fewer reductions on the CPU, **no time saved**

1.25×
faster complete solver on one V100

1.59×
finer grid admitted at four GPUs, $n = 337$ to $n = 535$

CONCLUSION

Communication avoidance pays only when the operator, the kernel and the machine all line up.

FUTURE WORK

A tile-aware R_v that predicts read redundancy from tile dimensions, stencil radius and streamed height, validated against the geometry sweeps already collected.

Mixed-precision recurrence with higher-precision Gram accumulation, gated on measured orthogonality and action error rather than on arithmetic intensity alone.

Newer fabric and more participants: H100-class devices and faster links would test whether the crossover transfers beyond syngé.

REFERENCES

- [1] J. Niesen and W. M. Wright, "Algorithm 919: A Krylov subspace algorithm for evaluating the phi-functions appearing in exponential integrators," ACM Trans. Math. Softw. 38(3), 2012.
- [2] M. Hoemmen, "Communication-avoiding Krylov subspace methods," Ph.D. thesis, University of California, Berkeley, 2010.

- [3] D. M. Dang, C. C. Christara, and K. R. Jackson, "A parallel implementation on GPUs of ADI finite difference methods for parabolic PDEs with applications in finance," SSRN, 2010.
- [4] A. H. Al-Mohy and N. J. Higham, "Computing the action of the matrix exponential, with an application to exponential integrators," SIAM J. Sci. Comput. 33(2):488-511, 2011.

